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Improvement of Erinacine A Productivity in *Hericium erinaceus* Mycelia and Its Neuroprotective Bioactivity Against The Glutamate-Insulted Apoptosis

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***Running title*:** *Hericium erinaceus* protected glutamate-insulted apoptosis.

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Abstract

Hericium erinaceus (Bulliard: Fries) Persoon (HE), synonymously called Monkey Head mushroom or Yamabushitake, contains a huge amount of nutrients, in particular, the erinacines. This study was carried out with the aim to uncover whether erinacines can play a role as a neuroprotective adjuvant and what the related mechanism of action could be. A novel medium was developed to significantly enhance the productivity of biomass as well as erinacines to reach 16.8 g/L and 225±54 mg/L, respectively. The ethanolic extract of HE (HEE) was tested on PC-12 cell line for its protective effect against the 20 mM glutamate-insult (GluI). The parameters affected were (control vs. GluI): glutathione (8.9±0.8 vs. 2.5±0.7 nmol/mg protein); glutathione peroxidase (11.6±1.6 vs. 28.2±3.2 mU/mg protein); glutathione reductase (9.2±0.8 vs. 2.3±0.4 mU/mg-protein); Ca^{2+}-influx (115±16 vs. 360±23 nmol/L); reactive oxygen species (100±3 vs. 140±7%); superoxide dismutase (56.5±6.3 vs. 23.2±4.2 U/mg-protein); H_2O_2 (8.2±0.7 vs. 20.4±3.5 nmol/ mg-protein); and TBARs (MDA) (8.1±0.3 vs. 13.3±2.5 μmol/μg-protein) (p<0.05). HEE effectively suppressed the ratio Bax/Bcl-2 and caspase-3, resulting in promising inhibition on PC-12 cell apoptosis induced by Glu-insult. Conclusively, the novel medium enhances both the productivity of biomass and erinacine A and its extract can effectively prevent PC-12 apoptosis *via* the ROS-caspase dependent pathway.

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Keywords *Hericium erinaceus*, erinacines, neuroprotective, glutamate-insult, glutathione peroxidase cycle.

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1. INTRODUCTION

Hericium erinaceus (*Bulliard: Fries*) *Persoon*, *Basidiomycetes* (HE), widely spread in China, Korea, Japan and North America, synonymously also is named Monkey Head mushroom (Chinese), Lion's Mane mushroom (Malaysian) and Yamabushitake (Japanese) (**Fig. 1**). For centuries HE has been used as a medicinal mushroom in Asia for the treatment of ulcers and stomach complaints due to its immuno-modulating, anticarcinogenic and antimicrobial efficacies (Kim, Son, Yun, & Han, 2010). In daily life, Monkey Head mushroom already is a favorite restaurant hot dish. Aqueous ethanolic extract of HE exhibited neurite outgrowth activity (Hazekawa, Kataoka, Hayakawa, Uchimasu, Furuta, & Fujiwara, 2010). Daily oral administration of aqueous extract of *H. erinaceus* fruiting bodies promoted the regeneration of injured rat peroneal nerve (Kim, Son, Yun, & Han, 2010; Wong, Naidu, David, Bakar, & Sabaratnam, 2012). *H. erinaceus* extract containing erinacine A, erinacine B, erinacine C and erinacine H promoted expression of neurogrowth factor (NGF) (Kawagishi, Shimada, Shirai, Okamoto, Ojima, & Furukawa, 1994; Lee, Shizuki, Hosokawa, Suzuki, Suganuma, & Kawagish, 2000; Mori, Obara, Hirota, Azumi, Kinugasa, & Nakahata, 2008; Shimbo, Kawagishi, & Yokogoshi, 2005). Functional deficiency of NGF has been considered to be related with Alzheimer's disease (Chang, Wang, Liaw, Peng, & Peng, 2012; Li, Larson, Sonnen, Shofer, Petrie, Schantz,

Montine, 2007). Human clinical tests revealed HEF to be effective in improving mild cognitive impairment (Mori, Inatomi, Ouchi, Azumi, & Tuchida, 2009). Literature elsewhere has indicated that oxidative stress and glutathione depletion could be the etiological factors inducing neurodegenerative cognitive impairment (Cai, Wang, Li, Qian, & Yao, 2008). Cultured rat pheochromocytoma cell line (PC-12 cell line) has been used extensively as an *in vitro* model for investigation of neuronal differentiation (Eik, Naidu, David, Wong, Tan, & Sabaratnam, 2012) and protection, for which glutamate and β-amyloid are two insulters frequently used (Chang, Wang, Liaw, Peng, & Peng, 2012). It was hypothesized that the neuroprotective effect of *H. erinaceous* not only could be assessed by the NGF stimulating effect, but also can be approached by the antiapoptotic and antioxidative pathways. To verify this, the HEF mycelia was selected to produce erinacines instead of using the fruiting bodies to avoid most of the nutritional interference coming from the fruiting bodies (Mau, Lin, Ma, & Song, 2001) and the PC-12 cell-glutamate model was used to carry out this present experiment.

2. MATERIALS AND METHODS

2.1 Chemicals

Unless otherwise stated, the chemicals and reagents used in this experiment were all supplied by Sigma-Aldrich Co. (St. Louis, MO, USA).

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2.2 Fungal strain

Hericium erinaceus (Bulliard: Fries) Persoon ATCC 56457 (DQ185920), also coded as BCRC 35669, was purchased from the Bioresources Collection And Research Center (BCRC) of Food Industry Research and Development Institute (Hsinchu, Taiwan).

2.3 *Mycelium production in a pilot scale bioreactor*

The cultivation method was conducted according to (Krzyczkowski, Malinowska, & Herold, 2010) with slight modification. In brief, the HE mycelia were first grown on potato dextrose agar (PDA, Difco). A piece of 20mm×20mm agar culture was inoculated into 1 L-broth consisting of (g/L): glucose 66, casein peptone 12, NaCl 1.45, KH_2PO_4 1.0; $MgSO_4$•$7H_2O$, 0.4; $FeCl_2$, 0.02; $CuSO_4$•$5H_2O$, 0.010; $ZnSO_4$ 0.055, $Mn(NO_3)_2$, 0.007; and $Ni(NO_3)_2$, 0.005 (denoted as cultivation medium, CM). The initial pH value was adjusted to 4.5 using 1 N HCl. The finished medium was transferred into a 2L-Hinton flasks. A total of ten 2L-Hinton cultures were prepared. The cultivation was carried out in a rotary shaker at 28 °C for 10 days with constant shaking at 100 rpm. The finished starter was transferred to a 500 L-bioreactor containing 400 L of CM. The cultivation was continued at 28 °C for 8 days with a constant stirring speed 30 rpm and an aeration rate 50 L/min. The mycelia were

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harvested by filtration, rinsed twice with deionized water, lyophilized, and weighed (HMC).

2.4 Extraction of erinacines

The HMC (10g) was extracted with 80% ethanol at 1:3(v/v) for isolation of erinacines as reported by Kawagishi, Shimada, Shirai, Okamoto, Ojima, & Furukawa (1994). The extract was repeated for three times. The extracts were combined and evaporated using a rotary evaporator to 100 mL (the ethanolic extract of erinacines from HE mycelia, HEE).

2.5 HPLC and LC/MS/MS analysis

Aliquots (10 μL) of HEE were analyzed using a series 1260 Infinity HPLC system (Agilent Technologies, Santa Clara, CA, USA) equipped with ChemStation software (version: B.04.03) and packed models of G1379B degasser, G1312B binary gradient pump, G1329B autosampler, G1316A column oven, and G1315D diode array detection system. The analytical column [Symmetry C-18 column, 150×2.3 mm, i.d.= 3 μm (Waters, MA, USA)] connected with a precolumn (C18(ODS) 4 mm×3.0 mm i.d. (Phenomenex Inc., Torrance, CA, U.S.A.) was installed in the column oven preset at 30 °C and operated at a flow rate of 0.2 mL/min for analyses of HEE. Gradient elution was applied using solvent A (water, containing 0.1% formic acid, FA) and solvent B (acetonitrile, containing 0.1% FA) as the mobile phase. The programmed elution was

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performed as follows: 0–5 min linear from 10% to 95% B; 50–65 min, isocratic at 95% B; finally, 10% B for 5 min. The absorption spectra of the eluted compounds were scanned within 200–400 nm using the inline diode array detectors (DAD) monitored at 254, 280, 320, and 350 nm, respectively for the UV–vis absorption. The compounds separated by elution were concomitantly identified using the Agilent 6420 Triple Quadrupole Mass Spectrometer equipped with a Mass Hunter software (version: B.01.04) (Agilent Technologies, Santa Clara, CA, USA).The system was operated in electrospray ionization (ESI) in the positive ionization modes under the following conditions. Nitrogen gas operated at a flow rate 10 L/min and a pressure 30 psi was used as both the drying- and the nebulizing gas. The drying temperature was held at 300 °C. A potential 3500 V was applied across the capillary. The fragmentor voltage was set at 90 V. In MS/MS analysis, the quadrupole 1 filters the calculated m/z of each compound of interest, whilst the quadrupole 2 scans for ions produced by nitrogen collision of these ionized compounds at a scan time 200 ms/cycle. The unknown compounds were identified by comparing with the cited data affiliated with the authorized UV–vis information and MS spectral data base.

2.6 Source and cultivation of PC12 cell line

Rat's adrenal pheochromocytoma cell line (PC12 cells) was obtained from the Bioresource Collection and Research Center of Food Industry Research and

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Development Institute (Hsin-Chu, Taiwan). The PC 12 cell line was cultivated with RPMI1640 culture medium as previously described (Chang, Wang, Liaw, Peng, & Peng, 2012).

2.7 The cytotoxicity and cell viability tests

The protective and viability experiment was tested in three separate categories: In experiment 1, the cytotoxicity of HEE on the viability of PC-12 cell line was conducted at different HE concentration (0.0, 50, 100, and 150 μg/mL). In experiment 2, the toxicity of glutamate (pH 7.0) to the viability of PC-12 cell line as described by (Kawakami, Kanno, Ikarashi, & Kase, 2011) was tested at concentrations 0.0, 1.0, 5.0, 10.0, 15.0, 20.0, and 25 mM, from which the optimum concentration was selected. In experiment 3, the protective effect of HE (0.0, 50, 100, and 150 μg/mL) against the insult provoked by glutamate at the selected concentration (20 mM) was explored. PC-12 cells were cultivated as previously cited (Chang, Chen, Chiou, Peng, & Peng, 2010; Chang, Wang, Liaw, Peng, & Peng, 2012). Briefly, PC-12 cells were seeded to each well at a density of 1×10^4cell/well and incubated a 37 ^oC under 5% CO_2 atmosphere. The cultivation was carried out at 37^oC under 5% CO_2 atmosphere until 80% confluent. The medium was changed with fresh media containing HEE 0.0, 50, 100, and 150 μg/mL, respectively. Medium only and medium containing 4% ethanol were used respectively as the controls. The protective effect of HEE on the cell

viability after being subjected to glutamate-insult was evaluated by 3-(4,5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) assay at 24 and 48 h, respectively as previously reported (Chang, Wang, Liaw, Peng, & Peng, 2012).

2.8 Protein extraction from the PC-12 cells

The PC12 cells in 24-well plates treated differently were harvested after 24 h incubation. The cells were centrifuged at 4°C at 2000×g for 10 min. The supernatant was decanted off. The cell pellets (CPL) were rinsed with PBS thrice. Lysis buffer (10 mL) was mixed with 100 μL of protease inhibitor and mixed well (protein inhibitor lysis buffer, PILB). To 200 mg of CPL, 1 mL of PILB was added. The mixture was homogenized and centrifuged at 4°C 14000×g for 20 min. The supernatant was separated and transferred to a new microcentrifuge tube. The mixture containing the whole intracellular proteins (WICP) was determined for its protein content using the Bradford and Lowry protein assay kits (BioRad, Hercules, CA., USA) by following the manufacturer's instructions. The separated proteins was stored at −80°C when not used or directly used for testing the content (or activity) of glutathione, glutathione peroxidase (GPX), glutathione reductase (GR), superoxide dismutase (SOD), and hydrogen peroxide as described in the following sections.

2.9 Determination of reduced glutathione (GSH) content

The glutathione content was assayed according to the method as previous reported

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(Hsieh, Wang, Tsai, Peng, & Peng, 2012). The authentic glutathione (Sigma-Aldrich Co., St. Louis, MO., USA) was used and treated similarly as mentioned to establish the calibration curve. From which the content of glutathione in PC-12 cells was obtained. The content of glutathione was expressed in nmol/mg protein

2.10 Assay for glutathione peroxidase (GPX)

Glutathione peroxidase activity in WICP was measured by following NADH oxidation at 340 nm as described by Flhé & Gunzler (1984). The reaction medium consisted of 30 mM phosphate buffer (pH adjusted to 7.0), 0.17 mm GSH, 0.2 U/mL glutathione reductase and 0.5 mM *tert*-butyl hydroperoxide. One unit corresponds to 1 μmol NADH per min per mg of protein. The GPX activity was expressed as mU/mg of protein.

2.11 Assay for glutathione reductase (GR)

The method of Mohandas, Marshall, Duggin, Horvath, & Tiller (1984) was followed to determine the activity of GR. The reaction mixture (1 mL) consisted of 0.1 M potassium phosphate (pH 7.6), 0.1 mM NADPH, 0.5 mM EDTA, 1 mM GSSG, and 100 to 200 μL of WICP (containing cytosol or microsomal suspension, 6 mg protein/mL). The reaction mixture was incubated at 30°C for 5 min before initiating the reaction. The enzyme activity was determined by measuring the disappearance of NADPH at 340 nm. One unit corresponds to 1 μmol NADH per min per mg of protein.

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The activity of GR was expressed as mU NADPH oxidized per min per mg of protein.

2.12 Determination of superoxide dismutase (SOD) activity

An aliquot of 100-200 μL of WICP was used for the determination of SOD using the Cayman's Superoxide Dismutase Assay Kits by following the instructions given by the manufacturer Cayman Chem. Co. (MI, USA). The optical density was read at 450 nm. Standard SOD was prepared and similarly treated as directed by the manufacturer. The activity of SOD was calculated and expressed in U/mg protein.

2.13 Determination of hydrogen peroxide

The method for assay of hydrogen peroxide (H_2O_2) was performed by following the instructions given by the manufacturer Cell Biolabs Inc. The amount of intracellular hydrogen peroxide was expressed in nmol/mg of protein.

2.14 Determination of the calcium influx

The determination of the intracellular calcium ion level was carried out as previously described (Chang, Wang, Liaw, Peng, & Peng, 2012). PC-12 cells at a density of 1×10^4 cells/well were plated onto a 6-well plate and incubated for 24 h. To each well, 1 μg/mL of ACEE was added, and the incubation was continued for 24 h. The chemifluorescence was measured at λ_{Ex} =488 nm and λ_{Em} =532 nm. Triplicate experiments were conducted and the data obtained were statistically treated.

2.15 Determination of the total intracellular reactive oxygen species (ROS) using

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2',7'-dichlorofluorescine diacetate

The protocol for determination of total intracellular ROS was carried out as previously described (Chang, Wang, Liaw, Peng, & Peng, 2012). Briefly, PC12 cells at a density of 1×10^6 cells/well were plated onto a 6-well plate and incubated for 24 h. To each well, 1 μg/mL of ACEE was added and the incubation was continued for an additional 24 h. To each well, 15 μM of $A\beta_{25-35}$ was added and incubated for 24 h. The culture was transferred onto a 96-well plate at a density of 2×10^4 cells/well. To each well, 10 μM DCFH-DA (Beyotime Institute of Biotechnology) was added and left to stand for 30 min avoiding direct sunlight. The chemifluorescence was measured at λ_{Ex} = 488 nm and λ_{Em} = 532 nm. The experiments were repeated in triplicate. The data obtained were treated statistically.

2.16 Determination of the thiobarbituric acid reactive substances (TBARs; or malondialdehyde, MDA)

The assay of TBARs was conducted according to the protocol as previously reported (Chang, Wang, Liaw, Peng, & Peng, 2012). Briefly, the PC12 cells in 24-well plates treated differently were harvested after 24 h incubation. The cells were centrifuged at 4°C at 1000×g for 10 min. The supernatant was decanted off. The authentic malondialdehyde (MDA) (Sigma-Aldrich) was similarly treated to establish the calibration curve. Alternatively, MDA reacts with thiobarbituric acid (TBA) to

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produce a compound exhibiting a molar extinguishing coefficient ε_{532} = 1.56× $10^5 \cdot M^{-1} \cdot cm^{-1}$ at 532 nm, from which the amount of MDA in the sample also can be accessed.

2.17 Western blotting

The protocol used to conduct Western blotting for anti-β-actin, anti-Bcl-2, anti-Bax, and anti-cleaved caspase 3 was carried out as previously described by (Chang, Wang, Liaw, Peng, & Peng, 2012). PC12 cells at a density of 1 × 10^4 cells/well were plated onto a 6-well plate and incubated for 24 h. To each well, 1 μg/mL of ACEE was added, and the incubation was continued for 24 h. Then $A\beta_{25\text{-}35}$ (15 μM) was added and incubated for an additional 24 h. The cells were harvested and homogenized with 1 mL of protein extraction solution (EDTA free) (iNtRON Biotechnology). After incubation on ice for 40 min, the homogenate was centrifuged at 12000 × g for 20 min at 4°C. The supernatant was collected. 50 μg of the lysate in PBS were boiled for 10 min. The boiled sample solutions were loaded onto a 7.5% SDS-PAGE, and the remaining procedures were carried out according to the manufacturer's instructions.

2.18 TUNEL assay

The protocol used to conduct the Terminal Deoxynucleotidyl Transferase-Mediated Biotinylated UTP Nick End Labeling (TUNEL) assay was carried out as previously described by (Chang, Wang, Liaw, Peng, & Peng, 2012). Terminal deoxynucleotidyl

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transferase-mediated biotinylated UTP nick end labeling (TUNEL) reaction was carried out according to the protocol given by the manufacturer (In situ cell death detection kit; Roche).

2.19 Statistical analysis

A completely randomised design was used as an experimental design, where different ethanol concentrations in the solvent were applied as treatments. The experimental data were subjected to one-way analysis of variance (ANOVA). The statistical significance of differences between mean values was established at $p < 0.05$, and the Duncan's new multiple range test was applied for all statistical analyses.

3. RESULTS

3.1 HPLC and LC/MS/MS analysis of HEE

The HPLC and LC/MS/MS analysis revealed five major compounds to be present in the ethanolic extract. Peak 4 was confirmed to be erinacine A, the other four components although were suspected to be erinacine-related yet unverified (Fig. 2).

3.2 Yield of biomass and erinacine A

The submerge culture method modified from (Krzyczkowski, Malinowska, & Herold, 2010) yielded higher productivity of HE mycelial biomass (16.8±0.4 g/L) and erinacine A (225±54 mg/L) (Table 1).

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3.3 Effect of glutamate-insult and HEE on cell viability

HEE up to 100 μg/mL not only was totally non toxic to PC 12 cells but also a stimulator for proliferation of PC 12 cells in a time- and dose–responsive manner (Fig. 3A). Conversely, glutamate-insult exerted a strong inhibitory effect on the cell viability in a gradually time- and dose-responsive fashion. The value of EC_{50} for glutamate was estimated to be 17 mM (Fig. 3B). Hence a dose of glutamate at 20 mM was used to carry out all the subsequent protective tests. HEE protected the cell viability against glutamate (20 mM)-insult in a dose-dependent manner (Fig. 3C).

3.4 Effect on the glutathione peroxidase-glutathione regeneration cycle

Glutamate activated glutathione peroxidase (GPX) and at the same time suppressed the activity of glutathione reductase (GR), leading to declined level of reduced glutathione (GSH). The changes in levels of GSH, GPX, GR and H_2O_2 were -71.9%, +143%, -75% and +248.8%, respectively (Table 2).

3.5 Oxidative stress induced by glutamate-insult and antioxidative capability of HE

Glutamate stimulated the calcium influx and ROS level to 360±23 nmol/L and 140±7 % compared to the control 115±16 nmol/L and 100±3%. HE efficiently alleviated these levels to 117±22 nmol/L and 103±5%, respectively (Table 2). The antioxidative parameters SOD was downregulated to 23.2±4.2 U/mg protein compared to the

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control 56.5±6.3 U/mg protein. HE effectively recovered the SOD level to 52.4±5.7 U/mg protein (Table 2). The levels of H_2O_2 and MDA were all substantially raised to 20.4±3.5 nmol/mg protein and 13.3±2.5 μmol/μg protein, which were effectively ameliorated by HE to 8.4±0.6 nmol/mg protein and 8.3±0.4 μmol/μg protein (Table 2), underlying the potent ameliorating effect of HE on these adverse oxidative stress induced by glutamate (Table 2).

3.6 Effect of glutamate and HE on the apoptotic ratio Bax/Bcl-2 and caspase 3

The apoptotic protein Bax was upregulated and simultaneously the antiapoptotic protein Bcl-2 was down regulated by glutamate-insult ($p< 0.05$) (Fig. 4A, Fig. B). The overall ratio Bax/Bcl-2 was approximately raised 1.2 fold over the control ($p< 0.05$) (Fig. 4C). Similarly, glutamate-insult upregulated caspase 3 to 117% compared to 100% of the control ($p< 0.05$) (Fig. 4D), indicating glutamate to be a strong apoptotic agent regarding PC 12 cells. HE at 100 μg/mL effectively rescued the apoptosis ($p<0.05$) (Fig. 4A - Fig. 4D). TUNEL assay apparently consolidated the glutamate-induced apoptosis of PC 12 cells and the rescue by HE (Fig. 5).

4. DISCUSSION

The biotechnology for producing HEF has been successfully developed in Taiwan using the so-called ‘Space Bags solid culture’. The productivity of erinacine A was

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found further improved by addition of certain essential minerals (Table 1). As well, the hot dish of 'monkey head fungus' also has become a favorite food in many restaurants (Fig. 1). Erinacines are strong stimulators for the biosynthesis of neurogrowth factor (NGF) (Kawagishi, Shimada, Shirai, Okamoto, Ojima, & Furukawa, 1994; Lee, Shizuki, Hosokawa, Suzuki, Suganuma, & Kawagish, 2000; Mori, Obara, Hirota, Azumi, Kinugasa, & Nakahata, 2008; Shimbo, Kawagishi, & Yokogoshi, 2005) and augmenter for catecholamine (Shimbo, Kawagishi, & Yokogoshi, 2005). Both NGF and catecholamines are crucial for proper neuronal function and maintenance. NGF induction correlates with increased nerve cell growth and neurite outgrowth and acts as a neurotrophic molecule required for nerve cell survival and repair (Lorz & Mehmet, 2009; Skaper, 2008; Zareen & Greene, 2009).

Nutritionally, *H. erinaceus* fruiting bodies contain huge amounts of myo-inositol, essential amino acids and various valuable mononucleotides (Mau, Lin, Ma, & Song, 2001). Recent interest in these compounds comes from the recognition that they may improve neurodegenerative-associated diseases such as Alzheimer's, age-associated dementia and cognitive decline. Biologically, the cytotoxicity of glutamate insult on PC12 cells results from protein sulfhydryl oxidation exclusive of the peroxidation of membrane phospholipids (Tyurin, Tyurina, Quinn, Schor, Balachandran, & Kagan, 1998). Apparently, HE could exert stronger protective effective on the early stage

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pathological change elicited by glutamate-insult (Fig. 3A - Fig. 3C), suggesting some early irreversible injuring effect of glutamate unable rescued by HE eventually (Fig. 3A - Fig. 3C).

On the other hand, GPX plays an important defensive role against the intracellular oxidative stress like H_2O_2 that usually induced neural cell death (Kim-Han & Sun, 1998). GPX4 rescued neurodegenerative diseases through its function in removal of lipid hydroperoxides. While GPX1 increased but GPX4 decreased in the Alzheimer's disease (AD) (Yoo, Gu, Xu, Kim, Carlson, Hatfield, 2010). The MDA level was highly raised in parallel to the upregulation of GPX in PC12 cells (Table 2), implicating the increased GPX activity attributed by GPX1. Among the five GPX isoenzymes, GPX1 always is considered to be the major enzyme responsible for removing H_2O_2 (Kelner, Bagnell, Uglik, Montoya, & Mullenbach, 1995; Mirault, Tremblay, Beaudoin, & Tremblay, 1991). Appropriate upregulation of GPX1 protects cells against oxidative damage and suppresses apoptosis induced by H_2O_2 (Kayanoki, Fujii, Islam, Suzuki, Kawata, & Taniguchi, 1996; Mills, 1957). However overexpression of cytosolic GPX1 could destroy glutathione peroxidase-glutathione regeneration cycle. A decrease in GSH levels can be observed in neurodegenerative diseases like PD and AD as well as schizophrenia (Drukarch, Jongenelen, Schepens, Langeveld, & Stoof, 1996; Lavoie, Chen, Dalton, Gysin, Cuenod, & Do, 2009).

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Literature also indicated that the manifestations of glutamate cytotoxicity in PC 12 cells involved increase in Ca^{2+} influx, ROS generation, H_2O_2 and MDA production (Ma, Liu, Jiao, Wang, Chen, Zhang, 2012).

Theoretically, the conversion of $\bullet O_2^-$ into H_2O_2 would be retarded when SOD activity is severely reduced. As a consequence, the direct toxicity resulting from H_2O_2-mediated damage would occur (Koo, Lee, Chung, Ko, & Kim, 2004). Alternatively, the influx of extracellular Ca^{2+} may en route L-type Ca^{2+} channels mediated by oxidative stress resulting from the depletion of GSH following the inhibition of GR (Doroshenko & Doroshenko, 2004) (Table 2). As well, Stridh et al. reported the GSH efflux is dependent on the concentration of Ca^{2+} influx (Stridh, Correa, Nodin, Weber, Blomstrand, & Sandberg, 2010). Literature elsewhere demonstrated that glutamate cytotoxicity in PC 12 cells was accompanied by an increment of Bax/Bcl-2 ratio and caspase-3 protein and the reduction of cell viability and mitochondrial membrane potential (MMP) (Ma, Liu, Jiao, Wang, Chen, Zhang, 2012). The decreased ratio Bax/Bcl-2 may implicate a substantially enhanced resistance to oxidative stress induced by Ca^{2+} (Kowaltowski, Vercesi, & Fiskum, 2000). Moreover, caspase-3 is a key executioner caspase involved in neuronal apoptosis and progression of Alzheimer's disease (Cai, Wang, Li, Qian, & Yao, 2008; Gorman, Szegezdi, Quigney, & Samali, 2005).

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5. CONCLUSION

The novel medium consisting of (in g/L) glucose 66, casein peptone 12, NaCl 1.45, KH_2PO_4 1.0, $MgSO_4 \cdot 7H_2O$ 0.4, $FeCl_2$ 0.02, $CuSO_4 \cdot 5H_2O$ 0.010, $ZnSO_4$ 0.055, $Mn(NO_3)_2$ 0.007, and $Ni(NO_3)_2$ 0.005 can improve the productivity of biomass and erinacine A after an 8 day-fermentative course, reaching 16.8 g/L and 225±54 mg/L, respectively which is slightly higher than that ever cited. HEE containing erinacine A exerts prominent antiapoptotic bioactivity for PC-12 cells against the glutamate-insult. The action mechanism of HEE is suggested to undertake the ROS-caspase dependent pathway, underlying the potential neuroprotective effect of Hericium erinaceus.

Conflict of Interest Statement

The authors declare that any potential conflict of interest has been disclosed before submission of this article to the Editor.

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Figure Legends

Fig. 1. Contemporary culture method of *Hericium erinaceus* and the 'Monkey Head Mushroom' hot dishes.

A. Solid culture on "Space Bags". B. Growing fruiting bodies *in situ*. C. desiccated mature *H. erinaceus* fruiting bodies. D. The delicious restaurant hot dishes of Monkey Head Mushroom.

Fig. 2. The HPLC and total ion chromatogram of ESI-(+)LC/MS of *H. erinaceum* extract.

HPLC was scanned with wavelength 200-400 nm (upper panel). ESI-(+)LC/MS was scanned with m/z 100-900 (lower panel). Five peaks were perceived. Peak 4 was identified to be erinacine A. Other four peaks remain unassigned.

Fig. 3. Cell viability tests.

A.Viability affected by HE (0.0, 50.0, 100.0, 100.0 μg/mL). B. Viability affected by glutamate (0.0, 1.0, 5.0, 10.0, 15.0, 20.0, and 25.0 mM). EC_{50} was estimated to be 17 mM. C. Viability rescued by HE in the presence of glutamate (20 mM). HE at 100 μg/mL rescued 90% of the injured PC 12 cells after insulted with glutamate (20 mM) for 24 h, and 85% at 48 h (refer to Fig. 4).

Different superscripts in upper case indicate significant difference after cultured for 24 h. Different superscripts in lower case indicate significant difference after cultured for 48 h.

Fig. 4. Western blot of apoptotic and antiapoptotic proteins affected by glutamate and HE.

A. Bax protein. B. Bcl-2 protein. C. ratio of Bax/Bcl-2. D. Caspase-3**.**

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Different superscripts in lower case indicate significant difference after cultured for 24h.

Fig. 5. TUNEL assay.

PC 12 cells were insulted with glutamate 20 mM and severe apoptosis was seen after insulted for 24 h. HE (100 μg/mL) rescued almost 80% of the apoptotic cells (refer to figure 2C).

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Table captions

Table 1. The fermentation parameters for production of erinacines by *H. erinaceous* mycelia[a]

Table 2. Biochemical parameters provoked by glutamate insult and the protective effect of *H. erinaceus* extract on PC-12 cells[a,b]

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Table 1. The fermentation parameters for production of erinacines by *H. erinaceous* mycelia[a]

Parameters	Starter[b]	Pilot bioreactor[b]	
		initial	final
Biomass[c], g/L	5.6±0.3	0.14±0.02	16.8±0.4
pH	4.5	4.5	3.6
Temperature, ^{o}C	28	28	28
Aeration rate, L/min	Free diffusion	50±0.3	50±0.3
Agitation speed, rpm	100	30	30
Erinacine A, mg/L	13.4±8	UD[d]	225±54

[a]The modified medium consisted of (g/L): glucose 66, casein peptone 12, NaCl 1.45, KH_2PO_4 1.0; $MgSO_4 \cdot 7H_2O$, 0.4; $FeCl_2$, 0.02; $CuSO_4 \cdot 5H_2O$, 0.010; $ZnSO_4$ 0.055, $Mn(NO_3)_2$, 0.007; and $Ni(NO_3)_2$, 0.005 (denoted as cultivation medium, CM).

[b]10 days in Hinton flasks agitated at 100 rpm in the rotary shaker. 8 days in the pilot bioreactor (working volume 400 L).

[c]On dry basis. Data calculated from triplicates (n= 3).

[d]UD: undetectable

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Table 2. Biochemical parameters provoked by glutamate insult and the protective effect of *H. erinaceus* extract on PC-12 cells[a,b]

Parameters[b]	Control	Glutamate, 20 mM	Glutamate 20 mM +HE 100 μg/mL[†]
GSH (nmol/mg protein)	8.9±0.8	2.5±0.7*	8.6±0.4
GPX (mU/mg protein)	11.6±1.6	28.2±3.2*	11.2±1.3
GR (mU/mg protein)	9.2±0.8	2.3±0.4*	7.8 ± 0.5
Ca^{2+}-influx (nmol/L)	115±16	360±23**	117±22
ROS (%)	100±3	140±7**	103±5
SOD (U/mg protein)	56.5±6.3	23.2±4.2**	52.4±5.7
H_2O_2 (nmol/ mg protein)	8.2±0.7	20.4±3.5**	8.4±0.6
TBARs (MDA), (μmol/μg protein)	8.1±0.3	13.3±2.5*	8.3±0.4

[a]Data are expressed in mean±SD (n = 6). Star symbols in the same row indicate statistically different vs. the control at *p<0.05 and **p<0.001.

[b]GSH, glutathione; GPX, glutathione peroxidase; GR, glutathione reductase; ROS, reactive oxygen species; SOD, superoxide dismutase; TBARs, thiobarbituric acid reactive substances; HE, the ethanolic extract of *H. erinaceous*.

Commercialized solid culture of *H. erinaceum* using "Space Bags" [1]	*Fruiting bodies* growing on Space Bags [1].
Desiccated mature fruit bodies of *H. erinaceum* [2]	Delicious restaurant hot dishes of Monkey Head Mushroom [3]

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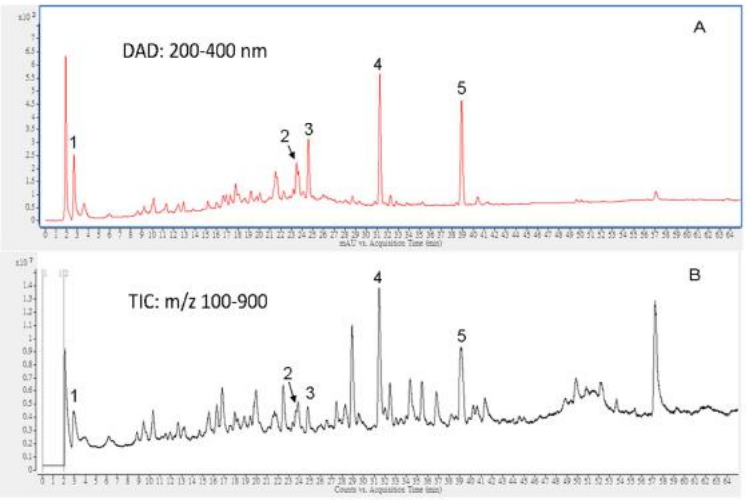

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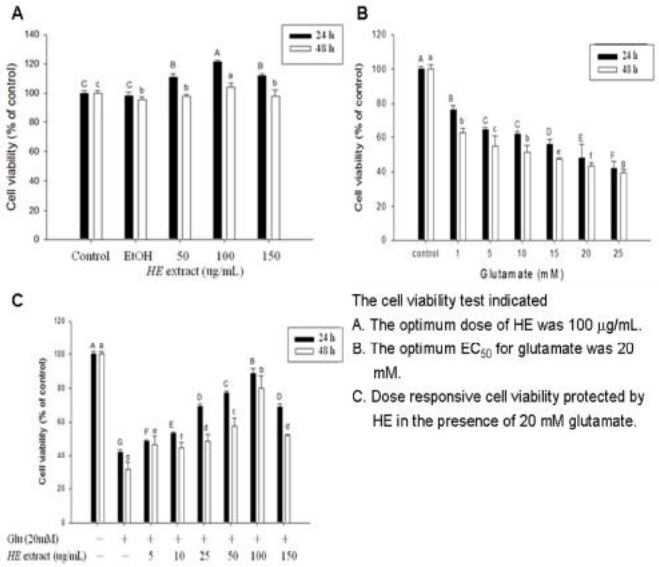

The cell viability test indicated

A. The optimum dose of HE was 100 μg/mL.

B. The optimum EC_{50} for glutamate was 20 mM.

C. Dose responsive cell viability protected by HE in the presence of 20 mM glutamate.

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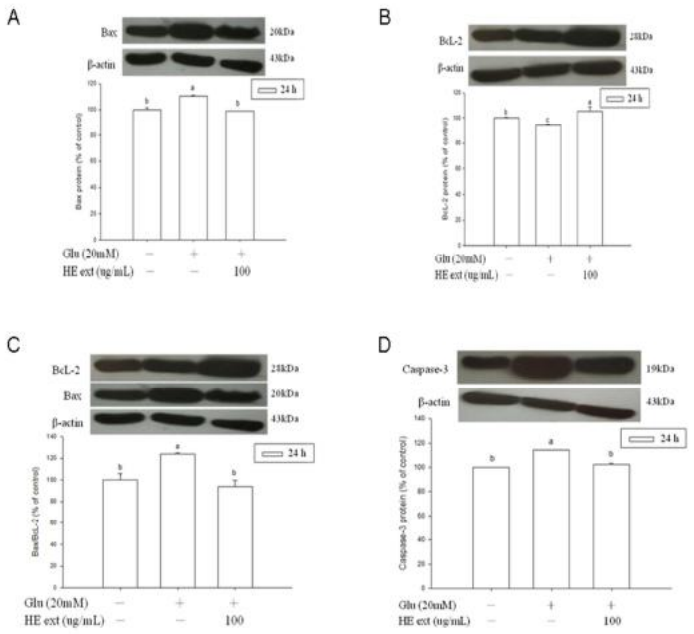

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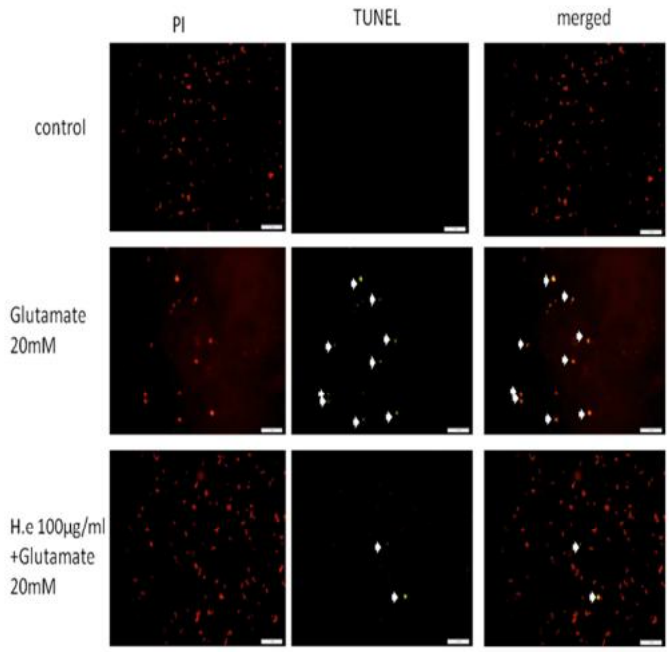

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Highlights

•Productivity of erinacines was improved in *Hericium erinaceus* (HE).

•Promising neuroprotective effect was found against glutamate insult (GluI).

•Parameters for alleviation associated with neuroprotection included glutathione,

•Glutamate-insulted apoptosis of PC 12 cells was secured.

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